Risk Assessment - 3560RA

Internal Reference: (CTP)

Information valid as of 20/05/2026 10:57

High Risk

SIGN-OFF REQUIRED

This assessment must be digitally signed off by an Authorised User.

General Details

Assessor	Samuel Prieto Ayllon
Assessment Date	18/05/2026
Assigned Reviewer	Samuel Prieto Ayllon
Next Review Date	18/05/2027

Assigned user to Sign Off	Oraib AlKheetanRojas
Signed Off By	This assessment has not yet been signed off
Signed Off On	--
Sign Off Comments	

Operation Assessed	Operation of Collaborative Robotic Arms in the KINESIS CTP Lab

Associated with specific area

- Basement 2
- B2 029 (Robotics Lab)
- CTP

Description of work area and/or activity assessed

This risk assessment covers the routine operation, programming, testing, demonstration, maintenance, and setup of collaborative robotic arms in the Kinesis CTP laboratory. Activities may include powering on/off, jogging or teaching the robot, running programmed movements, testing end-effectors, mounting payloads, connecting sensors or tools, working in manual or automatic modes, emergency stop testing, adjustment of workpieces, and post-use shutdown. The assessment applies to collaborative robotic arms used for research, teaching, prototyping, integration, and demonstration activities under controlled laboratory conditions.

Reason for Risk Assessment

To identify, assess, and control health, safety, collision, crushing, trapping, pinch-point, electrical, fire, payload, tooling, equipment-damage, and operational risks associated with the use of collaborative robotic arms during research, teaching, testing, demonstrations, and routine laboratory activities in the Kinesis CTP lab.

Overall Current Risk

High Risk

Average Number of Persons Affected

3	Cleaners
5	Employees
2	Machine Operators
3	Maintenance Staff
3	Trainees / Young Persons
2	Visitors

Hazards

Current Rating	Hazard Information	Task Status
Low Risk	**Electricity** Collaborative robotic arms use electrical power, control cabinets, power supplies, cables, sensors, end-effectors, and connected equipment. Damaged wiring, exposed terminals, incorrect connections, overloaded accessories, or unauthorised access to electrical components may cause electric shock, short circuit, burns, fire, equipment damage, or unexpected robot behaviour.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Only trained/authorised users operate, connect, or maintain the robotic arm. - Robot, controller, cables, plugs, end-effectors, and connected equipment are visually inspected before use. - Damaged cables, exposed conductors, loose connectors, or abnormal equipment conditions are reported and the system is removed from service. - Power is switched off before connecting/disconnecting end-effectors, payloads, sensors, or electrical accessories where practicable. - Manufacturer instructions are followed for powering, operation, connection, and shutdown. - Users do not open control cabinets or modify electrical systems unless authorised and competent. - Emergency stop and power-off methods are identified before operation begins.	

Potential Rating	Additional Controls Required
Low Risk	**No Remedial Actions Entered** If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.

Current Rating	Hazard Information	Task Status
High Risk	**Moving Parts** The robotic arm may move under manual, taught, programmed, remote, or automated control. Movement of the arm, joints, end-effector, payload, or workpiece may cause impact, trapping, pinching, crushing, or collision with operators, nearby personnel, equipment, benches, fixtures, or surrounding objects.	0 0 0
	Measures Currently in place to prevent risk of injury	
	- Only trained/authorised users operate or program the robotic arm. - Pre-use checks are completed before operation, including robot condition, end-effector, payload, work area, cables, and emergency stop. - Robot speed, force, payload, and operating mode are kept appropriate to the task and environment. - The robot operating area is kept clear of non-essential personnel and unnecessary objects. - Users keep hands, body parts, loose clothing, tools, and cables away from joints, end-effectors, and pinch points. - Movements are tested at reduced speed where practicable before normal operation. - Operators remain ready to stop the robot if unexpected movement occurs. - Emergency stop, protective stop, or power-off methods are available and tested/confirmed before work starts. - Work is stopped if the robot behaves unexpectedly or if anyone enters the operating area unsafely.	

Potential Rating	Additional Controls Required
High Risk	**No Remedial Actions Entered** If there are any reasonably practicable actions which can be taken to reduce the risk associated with this hazard please use the form above to enter them.